Load testing – CargoSync

Products:

For testing product endpoint, we have focused on the most critical endpoints. These include:

- GET /api/categories
- GET /api/products
- GET /api/products?name=Cola&minPrice=50
- GET /api/products?categoryName=...&page=0&size=20
- POST /api/products
- PUT /api/products/{id}

Category GET request is especially interesting since it has Redis cache. Product categories are very rarely changed, and are therefore and excellent candidate for caching.

Factories:

For factories, we chose these endpoints:

- GET /api/factories?cityCode=BEG
- GET /api/factories/{id}
- GET /api/factories/{id}/products/{id}/production
- GET /api/factories/{id}/products

Search is not being cached, and therefore offers a constant challenge to the backend. Meanwhile the ID get is, and let us see the optimization achieved from the caching. Production endpoint is great since it needs to generate influx queries and is therefore heavy for the influx database and the communication between it and the backend.

Users:

Lastly for users the tested endpoints are:

GET /api/managers?query=manager555

Login let us see how difficult it is for the server to handle the constant password hashing.

Results (users/ramping):

Test 200/10:

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	GET /api/categories	2993	0	1000	2300	3000	1164.89	3	4245	70	10.9	0
GET	GET /api/factories/{id}	2699	0	1500	2700	3300	1527.16	4	5197	277.3	13.2	0
GET	GET /api/factories/{id}/availability?6m	1889	0	1400	2600	3300	1417.67	31	4781	4658	7.3	0
GET	GET /api/factories/{id}/products	2241	0	1500	2600	3400	1491.77	6	4125	3382.99	10.2	0
GET	GET /api/factories/{id}/products/{id}/production?6m	1812	0	1400	2400	3200	1361.98	14	4953	2485.61	7.3	0
GET	GET /api/factories?cityCode=BEG	2656	0	1400	2600	3300	1387.87	9	5434	5411	11.5	0
GET	GET /api/products	2940	0	1600	2700	3500	1563.32	90	5408	783001.95	11.8	0
GET	GET /api/products?categoryName={name}	2172	0	1600	2700	3500	1538.22	92	4328	768301.59	10.2	0
GET	GET /api/products?name=Cola&minPrice=50	3022	0	1600	2700	3400	1550.18	84	5375	716865	13.1	0
POST	POST /api/products	764	0	1400	2500	3200	1404.8	7	3831	367.06	3.4	0
POST	POST /auth/login	1	0	68.1	68	68	68.1	68	68	377	0	0
PUT	PUT /api/products/{id}	2267	0	1600	2800	3400	1653.16	42	4712	10917.92	11.5	0

Test 400/20:

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	GET /api/categories	2008	0	2200	3800	4700	2309.03	4	6990	70	16.6	0
GET	GET /api/factories/{id}	1804	0	2800	4200	5000	2707.41	4	6335	277.33	14.8	0
GET	GET /api/factories/{id}/availability?6m	1291	0	2800	4000	4700	2577.78	29	6513	4658	9.5	0
GET	GET /api/factories/{id}/products	1430	0	2800	4200	5200	2719.43	6	9065	4925.73	13.6	0
GET	GET /api/factories/{id}/products/{id}/production?6m	1222	0	2700	3900	4500	2546.5	12	5279	2493.85	10.2	0
GET	GET /api/factories?cityCode=BEG	1775	0	2700	4000	4900	2544.23	10	6929	5411	13.1	0
GET	GET /api/products	2085	0	2800	4200	5300	2699.88	98	8150	753905.28	15.8	0
GET	GET /api/products?categoryName={name}	1542	0	2900	4100	4800	2711.3	83	7002	763910.86	11.9	0
GET	GET /api/products?name=Cola&minPrice=50	2012	0	2900	4200	5100	2749.41	98	7271	716875.65	16.2	0
POST	POST /api/products	528	0	2700	3800	4400	2554.95	14	8352	367.08	4.2	0
POST	POST /auth/login	1	0	65.96	66	66	65.96	66	66	377	0	0
PUT	PUT /api/products/{id}	1447	0	2900	4200	5000	2812.4	22	7376	11367.59	9.2	0

Test 600/30:

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	GET /api/categories	1961	29	2400	3600	30000	2751.51	3	30054	68.96	21	0
GET	GET /api/factories/{id}	1629	24	2700	4000	30000	3034.66	4	30055	273.07	15.4	0
GET	GET /api/factories/{id}/availability?6m	1216	9	2700	3900	4800	2723.5	41	30048	4623.52	10.2	0
GET	GET /api/factories/{id}/products	1433	17	2800	4000	30000	2958.97	6	30052	6202.92	15.4	0
GET	GET /api/factories/{id}/products/{id}/production?6m	1148	9	2600	3700	4700	2667.89	12	30052	2461.69	11.9	0
GET	GET /api/factories?cityCode=BEG	1666	14	2600	3700	4800	2694.26	8	30044	5365.53	17.6	0
GET	GET /api/products	1926	21	2800	3900	30000	2914.35	73	30050	693290.87	19.3	0
GET	GET /api/products?categoryName={name}	1384	9	2800	3800	4800	2786.38	40	30048	745884.79	14.3	0
GET	GET /api/products?name=Cola&minPrice=50	1889	26	2800	4000	30000	2993.83	85	30057	707008.99	15.3	0
POST	POST /api/products	479	6	2700	3800	30000	2897.52	7	30037	362.37	4.6	0
POST	POST /auth/login	1	0	63.67	64	64	63.67	64	64	377	0	0
PUT	PUT /api/products/{id}	1363	22	2700	4000	30000	2966.14	6	30046	347.19	14	0
GET	bootstrap categories	1	0	3	3	3	3	3	3	70	0	0
GET	bootstrap factories	1	0	22.09	22	22	22.09	22	22	27913	0	0
GET	bootstrap factory products	15	0	6	6	6	5.68	5	6	6184.73	0	0
GET	bootstrap products	1	0	308.99	310	310	308.99	309	309	3778673	0	0
	Aggregated	16113	186	2700	3900	30000	2854.74	3	30057	231770.95	159	0

Test 500/25:

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	GET /api/categories	3692	0	2800	4400	5500	2828.95	6	7224	70	19.5	0
GET	GET /api/factories/{id}	3159	0	3000	4700	5600	3113.9	7	7904	277.03	15.3	0
GET	GET /api/factories/{id}/availability?6m	2254	0	3000	4500	5400	2951.37	49	7277	4658	12.2	0
GET	GET /api/factories/{id}/products	2703	0	3000	4700	5600	3124.85	12	7209	7835.7	14.3	0
GET	GET /api/factories/{id}/products/{id}/production?6m	2312	0	2900	4400	5400	2909.98	18	8522	2494.13	11.7	0
GET	GET /api/factories?cityCode=BEG	3149	0	2900	4500	5500	2960.19	23	7912	5411	17.4	0
GET	GET /api/products	3625	0	3100	4800	5600	3121.5	82	7438	741533.95	18.8	0
GET	GET /api/products?categoryName={name}	2808	0	3100	4800	5800	3146.4	92	7331	757163.33	15.4	0
GET	GET /api/products?name=Cola&minPrice=50	3603	0	3100	4700	5700	3094.82	118	8085	716876	16.9	0
POST	POST /api/products	930	0	3000	4600	5600	3047.63	54	6636	367	3.5	0
POST	POST /auth/login	1	0	96.31	96	96	96.31	96	96	377	0	0
PUT	PUT /api/products/{id}	2726	0	3000	4700	5600	3052.33	17	7236	348.32	13.2	0
GET	bootstrap categories	1	0	16.42	16	16	16.42	16	16	70	0	0
GET	bootstrap factories	1	0	46.4	46	46	46.4	46	46	27913	0	0
GET	bootstrap factory products	15	0	5	11	11	5.9	5	11	7450.07	0	0
GET	bootstrap products	1	0	255.67	260	260	255.67	256	256	3814825	0	0
	Aggregated	30980	0	3000	4600	5600	3030.54	5	8522	240734.13	158.2	0

Conclusion:

From the results we can see that universally all endpoints started to fail at around 600 users. There are sevelar key factors to consider:

- Tests are being ran on a single machine. Specs are:
  - Windows 10 Laptop
  - Cpu - Intel Core i5 13500H
  - Gpu – NVIDIA GeForce RTX 4050 Laptop GPU
- Machine is using windows and is running docker stack via WSL. This offer significant limitation. Most notably it makes communication between the host system and container much slower and more demanding. It also limits the number of workers nginx can start.

Possible improvements:

GET /api/categories

Categories endpoint is already higly optimized. It uses caching for all entries, which allow fast reading.

GET /api/products

This endpoint results in getting the first page of the products. We don't have caching here, which causes it rerun the query every time. Since it's by far the most searched query, it's a good place for optimization.

GET /api/products?name=Cola&minPrice=50

This isn't a good place for caching, since queries are too dynamic. However, since it includes sorting, adding a database index can offer a significant boost in search performance.

POST /api/products

There is not much to improve when it comes to this endpoint. Backend uses optimistic locking which is necessary to keep the data integrity, even though it slows down calls.

GET /api/factories?cityCode=BEG

Since this is a query, with a simple query parameters, it is debatable whether it's a good idea to cache. I would rather not cache it due to high variance. However, it's still a very good place to add a database index for more optimized searching.

GET /api/factories/{id}

This endpoint uses cached data, which offers great performance. Since main data of factories rarely changes, it's an excellent place to put caching.

GET /api/factories/{id}/products/{id}/production

Highly demanding endpoint. Even though it's not included, a good consideration would be adding caching that lasts about 5 minutes. This would allow frequent requests to be delivered swiftly, while still keeping the data accurate.

GET /api/factories/{id}/products

This is also cached, and is therefore very optimized. There are no further improvements with the current tech stack.

GET /api/managers?query=manager555

Search query would highly benefit from database index. This would greatly improve the performance of this endpoint.